

TrajScale: Decoupled Mixed-Resolution Sampling for Efficient High-Resolution Video Generation

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Abstract

High-resolution video diffusion spends every denoising evaluation on a long target-resolution latent sequence, even though early evaluations primarily establish global structure and motion. Mixed-resolution sampling can move this prefix to a cheaper grid, but an intermediate resolution transition is not a simple resize: fixed interpolation ignores the cross-resolution representation induced by the video VAE, whereas a learned upsampler trained on converged clean latents receives a mismatched, unfinished prediction at deployment. We introduce **TrajScale**, which separates these two problems. A **Cross-Resolution Latent Upsampler (CRLU)** learns the model-specific clean-latent mapping across grids, while a transition-local **Endpoint Alignment Adapter (EAA)** calibrates the intermediate low-resolution prediction before CRLU is applied. A parameter-free, scheduler-consistent initialization then connects the upsampled clean latent to a short frozen high-resolution suffix, which synthesizes details not uniquely determined by the low-resolution input. Both learned components use model-endogenous supervision from the frozen target generator, without external paired videos or an auxiliary super-resolution teacher. On Wan2.1 with 50-step sampling and a $368 \times 640 \rightarrow 720 \times 1248$ transition, TrajScale-Q and TrajScale-E achieve $1.83\times$ and $2.22\times$ end-to-end speedups while retaining 97.76% and 97.53% of Native-HR VBench-5. Relative to trilinear interpolation, CRLU reduces latent L_1 error by 48.6% and LPIPS by 73.3%; EAA reduces the low-resolution endpoint residual by 25.8% and 21.0% at the two transition points and exposes a controllable detail–temporal-stability trade-off. TrajScale also transfers to a 4-step distilled sampler and shows that spatial and temporal acceleration are complementary.

1 Introduction

Recent latent video diffusion models such as Wan generate semantically coherent videos with realistic motion and fine spatial detail (Wan et al. 2025), but their inference cost grows aggressively with spatial resolution, video length, and sampling steps. In DiT-based generators, every denoising evaluation processes the full spatiotemporal token sequence. Executing the entire trajectory at the target resolution therefore forces early evaluations—which primarily establish global structure and macro-motion—to incur the same high-resolution cost as later evaluations that increasingly refine local structure and texture. Mixed-resolution inference offers

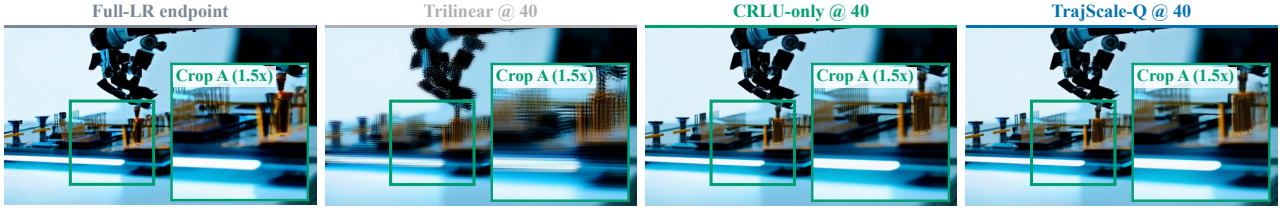
a complementary form of acceleration: a low-resolution prefix models global content economically, followed by a short high-resolution suffix that completes spatial refinement. This spatial strategy (ModelTC 2026; Zheng et al. 2026) is orthogonal to temporal step distillation and can be combined with few-step samplers.

Existing routes to higher-resolution video generation leave a gap between post-generation upscaling and in-trajectory acceleration. Conventional video super-resolution receives an already completed low-resolution RGB video and must independently recover underdetermined high-frequency content (Zhou et al. 2024; Xu et al. 2024). Cascaded RGB diffusion can reintroduce a generative prior, but decoding the low-resolution result and re-encoding it for a later stage adds codec overhead, while the completed RGB video has no intrinsic correspondence to a particular timestep or noise level of the high-resolution trajectory. Latent cascades avoid the RGB round trip, yet still require a manually chosen trajectory entry point and refinement strength after low-resolution generation has finished. In-trajectory methods avoid this late re-entry by resizing latents during sampling, but fixed interpolation does not learn the VAE-induced correspondence between two latent grids. As the transition is delayed for greater acceleration, fewer high-resolution evaluations remain to correct interpolation error. What is missing is a learned latent upsampler that can be inserted before low-resolution convergence without inheriting the transition-time residual of an unconverged trajectory state.

We propose **TrajScale**, a decoupled mixed-resolution sampling framework organized around this requirement. Its primary cross-resolution capability is a **Cross-Resolution Latent Upsampler (CRLU)** from model-generated paired clean latents, replacing geometry-only interpolation with a generator- and VAE-specific cross-resolution mapping. CRLU is designed to transfer structure and motion faithfully to the target latent grid rather than uniquely hallucinate every missing high-frequency detail; the frozen high-resolution suffix retains that generative responsibility. Deploying CRLU before the low-resolution trajectory has converged, however, presents it with a timestep-dependent input residual that is absent from its clean-latent training pairs. We therefore introduce a handoff-local **Endpoint Alignment Adapter (EAA)** that calibrates the transition-time clean prediction toward the endpoint of the corresponding native low-resolution trajec-

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(a) Matched real video frames — prompt 05, seed 9705, frame 41



(b)

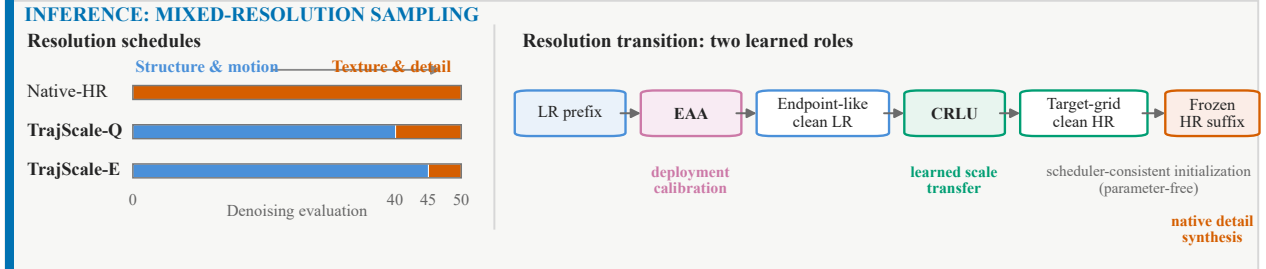


Figure 1: Real-frame comparison and TrajScale inference. (a) Content-aligned Full-LR endpoint, Trilinear, CRLU-only, and TrajScale-Q outputs for prompt 05, seed 9705, frame 41, with identical $1.5\times$ crops. The reference follows a complete 368p trajectory; transition outputs are 720p. (b) Native-HR performs all 50 evaluations at high resolution, whereas TrajScale-Q and TrajScale-E execute the first 40 and 45 evaluations at low resolution and retain 10 and 5 high-resolution evaluations. The transition contains two learned roles—EAA calibration and CRLU scale transfer—followed by parameter-free scheduler initialization and a frozen high-resolution suffix.

tory. The aligned latent is then upsampled, reconstructed at the corresponding scheduler level, and refined by the frozen generator through the remaining high-resolution evaluations. Thus, CRLU answers *how to change latent scale*, EAA answers *how to make an intermediate prediction suitable for that learned mapping*, and the frozen suffix remains responsible for generative detail completion.

Our primary contributions are summarized as follows:

- **We propose TrajScale**, a lightweight mixed-resolution video sampling framework that combines an inexpensive low-resolution prefix with a short high-resolution suffix, providing controllable quality–efficiency operating points and compatibility with temporally distilled samplers.
- **We develop CRLU**, a model-endogenous clean-latent up-sampler that learns the target generator’s cross-resolution VAE correspondence and consistently outperforms fixed latent interpolation, while delegating underdetermined high-frequency completion to the frozen high-resolution suffix.
- **We identify and calibrate the deployment mismatch of learned latent upsampling** through a transition-local EAA trained with endpoint supervision and cached-prefix consistency. Experiments on standard and 4-step Wan samplers show that CRLU supplies the principal cross-resolution gain, while EAA provides a targeted and controllable detail–stability adjustment at the transition.

2 Related Work

2.1 Video Super-Resolution and Cascaded Refinement

Latent Diffusion Models move generation into a compressed representation for scalable high-resolution synthesis (Romach et al. 2022). Conventional video super-resolution instead treats a completed low-resolution RGB video as the input to a separate restoration system (Zhou et al. 2024; Xu et al. 2024). Such a model must preserve structure and temporal consistency while also recovering high-frequency content that is not uniquely determined by the input. TrajScale does not formulate high-resolution generation as an isolated restoration problem: its learned scale operator transfers a stable latent representation, and the original generator remains responsible for underdetermined detail synthesis.

Cascaded generative systems, including latent-video pipelines and recent multi-stage systems such as LTX-2 and LUVE, improve resolution through a later upsampling or refinement stage (Blattmann et al. 2023; HaCohen et al. 2026; Zhao et al. 2026). RGB cascades incur an additional decode–encode round trip between stages and require a chosen noise level for the subsequent diffusion process. Latent cascades avoid the RGB round trip but still perform scale conversion after low-resolution completion and must determine where the upsampled latent should enter the high-resolution schedule. TrajScale instead inserts learned clean-latent upsampling before low-resolution completion and continues refinement within the frozen generator’s native high-resolution schedule.

2.2 In-Trajectory Mixed-Resolution Sampling

LightX2V changes latent resolution during sampling using fixed interpolation (ModelTC 2026). RALU shows that training-free latent upscaling can suffer from high-frequency aliasing and resolution-dependent noise–timestep mismatch and develops mixed-resolution strategies for Diffusion Transformers (Jeong et al. 2025). For image flow matching, MrFlow combines low-resolution structure formation with low-noise high-resolution refinement (Zheng et al. 2026). These methods establish the efficiency potential of in-trajectory resolution changes, but fixed resizing only changes geometry; it does not learn the cross-resolution correspondence induced by a specific video VAE. TrajScale replaces interpolation with CRLU. Because a learned clean-latent upsampler is trained on converged latent pairs, applying it to an unconverged trajectory prediction introduces a distinct training–inference input mismatch, which motivates EAA.

2.3 Trajectory Consistency and Few-Step Sampling

Temporal acceleration methods reduce the number of denoising evaluations via trajectory regression, consistency mapping, or distribution matching (Luo et al. 2023; Yin et al. 2024; Zhai et al. 2024). Recent approaches such as Self-Forcing and AnyFlow further emphasize consistency between training states and actual student rollouts to reduce trajectory distribution shift (Huang et al. 2025; Gu et al. 2026). TrajScale targets spatial rather than temporal computation. EAA is not a general low-resolution step-reduction model: it is activated only at the selected transition, while the preceding prefix is generated entirely by the frozen sampler. This locality yields the cached-prefix consistency property established in Section 3.3, allowing transition states to be reused without a student-prefix rollout mismatch.

3 Method

3.1 Two Mismatches in an Intermediate Resolution Transition

Let $v_\phi(x_k, k, c)$ denote a frozen Wan flow predictor, where x_k is the latent state before the k -th denoising evaluation, c is the text condition, and T is the total number of evaluations. Superscripts L and H denote the low- and target-resolution latent spaces. For a transition after evaluation s , x_s^L is the low-resolution state presented to the transition procedure. The frozen predictor produces an unaligned clean estimate

$$\hat{z}_s^L = x_s^L - \sigma_s v_\phi(x_s^L, s, c),$$

where σ_s is the scheduler coefficient at the transition. We use z_{end}^L for the endpoint obtained by completing the same native low-resolution trajectory under matched prompt, seed, initial noise, scheduler, and guidance. Separately, $(z_{\text{pair}}^L, z_{\text{pair}}^H)$ denotes a pair of clean Wan-VAE latents encoded from low- and high-resolution versions of the same generated video. The distinct notation is important: the native trajectory endpoint and the paired-latent training input are both clean-like, but they arise from different conditional distributions.

The handoff contains two different mismatches that should not be assigned to a single learned mapper. The *scale*

mismatch lies between z_{pair}^L and z_{pair}^H : fixed interpolation changes tensor geometry but does not model how the frozen VAE represents the same video on two latent grids. We address this mismatch with a clean-latent upsampler U_ψ trained on paired latent examples. The *deployment mismatch* appears only when U_ψ is moved from a completed latent cascade into an unfinished sampling trajectory. At transition s , $\hat{z}_s^L$ contains a step-, scheduler-, and guidance-dependent residual relative to the corresponding native endpoint. EAA reduces this residual before CRLU is applied.

These roles define the distributional bridge

$$p_{\text{handoff}}^{(s)}(\hat{z}_s^L) \xrightarrow{\text{EAA}} p_{\text{end}}(z_{\text{end}}^L) \longrightarrow p_{\text{pair}}(z_{\text{pair}}^L) \xrightarrow{\text{CRLU}} p_{\text{pair}}(z_{\text{pair}}^H).$$

The unlabeled middle arrow is deliberate. Native low-resolution endpoints and RGB-downsampled paired latents are both clean-latent distributions, but they are not identical; EAA is supervised only to reduce the transition residual, not to claim full distribution matching to CRLU’s training inputs. Although EAA precedes CRLU during inference, we present CRLU first below because the scale operator defines the capability, whereas endpoint alignment is required only by its intermediate deployment.

The resulting inference path is

$$x_s^L \xrightarrow{\text{EAA}} \hat{z}_s^L \xrightarrow{U_\psi} \hat{z}_s^H \xrightarrow{\text{parameter-free schedule initialization}} x_{s+1}^H \xrightarrow{\text{frozen HR suffix}} z_{\text{out}}^H$$

Here, $\hat{z}_s^L$ is the endpoint-aligned clean estimate and $\hat{z}_s^H$ is the CRLU output on the target grid. The two learned problems are

$$\mathcal{E}_{\text{scale}} = d(U_\psi(z_{\text{pair}}^L), z_{\text{pair}}^H), \quad \mathcal{E}_{\text{align}}(s) = d(\hat{z}_s^L, z_{\text{end}}^L).$$

These objectives are measured in different domains and are not treated as additive components of a single transition loss. The remaining high-resolution budget $R = T - s$ is a third, non-learned operating condition: a later transition reduces computation but leaves fewer frozen high-resolution evaluations to synthesize details that the low-resolution input cannot uniquely determine.

3.2 Model-Endogenous Cross-Resolution Latent Upsampling

Why learned latent upsampling. A spatial interpolation rule can resize a tensor, but it cannot learn the VAE-induced correspondence between the source and target latent grids. CRLU is therefore trained to transfer the low-frequency structure, motion trajectory, and temporally coherent content of a clean latent into the target-resolution representation. Unlike conventional video super-resolution, its objective is not to uniquely reconstruct every missing high-frequency texture. Those degrees of freedom remain available to the frozen generator during the high-resolution suffix.

Generator-endogenous scale supervision. We use the frozen Wan model to generate high-resolution reference videos from predefined prompts and random seeds. Each video is spatially downsampled in RGB space, after which the high- and low-resolution versions are independently encoded by the same frozen Wan VAE. This produces aligned pairs $(z_{\text{pair}}^L, z_{\text{pair}}^H)$ with identical frame counts, semantics,

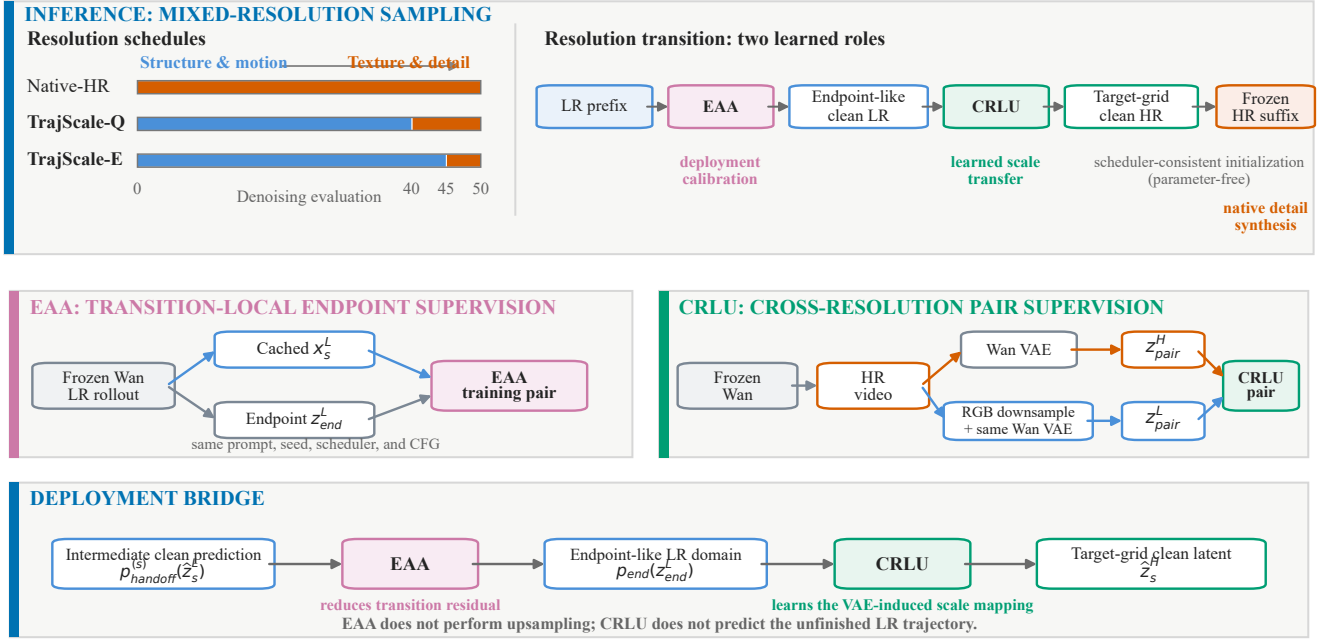


Figure 2: TrajScale overview. The inference path contains two learned roles: EAA calibrates the transition-time prediction toward its matched native low-resolution endpoint, and CRLU transfers the calibrated clean latent to the target grid. Scheduler-consistent initialization is parameter-free; the frozen high-resolution suffix performs detail synthesis. EAA pairs come from a cached state and the endpoint of the same frozen low-resolution rollout. CRLU pairs come from independent Wan-VAE encodings of a generated high-resolution video and its RGB-downsampled counterpart. The bottom row makes explicit that EAA reduces a deployment residual but does not perform spatial upsampling.

and motion trajectories. Performing the resolution change before VAE encoding is essential: CRLU learns the actual correspondence induced by the target generator, RGB down-sampling process, and VAE rather than imitating an arbitrary interpolation rule inside latent space.

Spatiotemporal latent upsampling. CRLU maps a 16-channel Wan latent to a target-resolution latent while preserving the temporal dimension:

$$U_\psi : \mathbb{R}^{B \times 16 \times F \times h \times w} \rightarrow \mathbb{R}^{B \times 16 \times F \times H \times W}.$$

Inspired by the latent upsampler in LTX-2 (HaCohen et al. 2026), the network follows an encoder-resampler-decoder design. A 3D convolution projects the input into a 256-dimensional hidden representation, followed by four spatiotemporal residual blocks. A spatial resampler changes only height and width; four additional 3D residual blocks and a final 3D convolution reconstruct the 16-channel output. The temporal dimension remains unchanged, while the 3D blocks allow adjacent latent frames to contribute to reconstruction.

We instantiate the spatial resampler for each latent-grid ratio. For $480 \times 832 \rightarrow 720 \times 1248$, the grid changes from 60×104 to 90×156 . CRLU expands channels by a factor of nine, applies a $3 \times$ spatial PixelShuffle, and performs stride-2 downsampling with a fixed binomial blur kernel:

$$60 \times 104 \xrightarrow{\times 3} 180 \times 312 \xrightarrow{/2} 90 \times 156.$$

For $368 \times 640 \rightarrow 720 \times 1248$, the grid changes from 46×80 to 90×156 . A $2 \times$ PixelShuffle produces a 92×160 grid, which

is center-cropped to the target size. These resolution-specific operators implement the required rational scale changes, while the surrounding spatiotemporal network learns the latent correspondence.

Cross-resolution objectives. Given $\hat{z}_{\text{pair}}^H = U_\psi(z_{\text{pair}}^L)$, CRLU is trained with

$$\mathcal{L}_{\text{CRLU}} = \mathcal{L}_{\text{rec}} + \lambda_{\text{content}} \mathcal{L}_{\text{content}} + \lambda_{\text{temp}} \mathcal{L}_{\text{temp}},$$

where

$$\mathcal{L}_{\text{rec}} = \mathbb{E} \left[\sqrt{(\hat{z}_{\text{pair}}^H - z_{\text{pair}}^L)^2 + \epsilon^2} \right],$$

and

$$\mathcal{L}_{\text{content}} = \|D(\hat{z}_{\text{pair}}^H) - z_{\text{pair}}^L\|_1, \quad (1)$$

$$\mathcal{L}_{\text{temp}} = \|\Delta_F \hat{z}_{\text{pair}}^H - \Delta_F z_{\text{pair}}^H\|_1. \quad (2)$$

Here, D maps the prediction back to the low-resolution grid, and Δ_F denotes first-order temporal differences between adjacent latent frames. We use $\lambda_{\text{content}} = 0.2$, $\lambda_{\text{temp}} = 0.1$, and $\epsilon = 10^{-3}$. The three terms constrain target-latent reconstruction, low-frequency content preservation, and temporal evolution, respectively.

A low-resolution latent does not uniquely specify every texture in its high-resolution counterpart. CRLU therefore produces a structurally faithful and temporally stable clean latent on the target grid, while the frozen high-resolution suffix supplies the generative degrees of freedom absent from the input. This division distinguishes TrajScale from a standalone super-resolution model that must recover all missing detail by itself.

3.3 Transition-Time Endpoint Alignment

Deployment mismatch. CRLU is trained on paired clean latents, but mixed-resolution acceleration presents it with the clean prediction of an unconverged trajectory state. In general,

$$p_{\text{handoff}}^{(s)}(\tilde{z}_s^L) \neq p_{\text{pair}}(z_{\text{pair}}^L).$$

The discrepancy contains a transition-specific residual whose magnitude and structure depend on the handoff step, scheduler, and guidance configuration. Passing this residual directly through CRLU entangles scale conversion with unfinished low-resolution denoising. EAA is therefore trained for CRLU deployment rather than as a general-purpose low-resolution acceleration model: it moves the transition-time prediction toward an endpoint-like clean-latent domain before scale conversion. It reduces, but does not claim to eliminate, the remaining gap between native trajectory endpoints and CRLU’s paired-latent training distribution.

Handoff-local endpoint supervision. For each training condition, we run the frozen model with a fixed text prompt, random seed, initial noise, scheduler, and Classifier-Free Guidance (CFG) configuration. We cache x_s^L immediately before the designated transition and continue the same native low-resolution trajectory to obtain z_{end}^L . The pair $(x_s^L, z_{\text{end}}^L)$ preserves the exact semantic and stochastic conditions of the handoff while isolating the residual that would otherwise be passed to CRLU.

Localized low-rank adaptation. EAA is implemented with LoRA (Hu et al. 2022) and is activated only at the designated transition step. The base Wan parameters remain frozen. For a linear layer W , the transition-time adapted weight is

$$W'(\gamma) = W + \gamma \frac{\alpha}{r} BA,$$

where r is the LoRA rank and $\gamma \geq 0$ is the inference-time alignment intensity. Thus, $\gamma = 0$ recovers the frozen predictor and $\gamma = 1$ applies the full learned update. Low-rank updates are injected into the q , k , v , and o attention projections and the `ffn.0` and `ffn.2` linear layers. Under the flow-matching parameterization, EAA produces

$$\tilde{z}_s^L = x_s^L - \sigma_s v_{\phi+\gamma\Delta\theta_s}(x_s^L, s, c).$$

We optimize

$$\mathcal{L}_{\text{EAA}} = \|\tilde{z}_s^L - z_{\text{end}}^L\|_1 + 0.1\|\tilde{z}_s^L - z_{\text{end}}^L\|_2^2.$$

For earlier transitions, where frame-wise residuals are more likely to propagate through scale conversion, we additionally apply a temporal residual alignment term with weight 0.05. Although the same target can be expressed as a velocity label $v^* = (x_s^L - z_{\text{end}}^L)/\sigma_s$, all reported models are optimized in the clean-latent domain. Training learns the adapter update; validation selects γ for each operating point. EAA performs no spatial upsampling, and increasing γ is not assumed to improve generation monotonically. Its sole role is to control how strongly the selected transition is moved toward the matched endpoint before CRLU is applied.

Cached-prefix consistency. Because EAA is inactive before step s , its low-rank parameters satisfy $\Delta\theta_k = 0$ for all $k < s$. Under deterministic sampling and matched text embeddings, initial noise, scheduler, CFG scale, and other conditioning inputs, the adapted and frozen processes therefore use identical transition functions throughout the prefix. By induction,

$$x_k^{\text{adapted}} = x_k^{\text{base}}, \quad \forall k \leq s.$$

Consequently, a prefix state cached from the frozen model is exactly the state consumed by EAA. This property supports offline pair construction and controlled comparisons among transition operators without introducing prefix rollout mismatch. The equality requires transition-local activation, deterministic prefix sampling, and matched schedulers, guidance, and conditioning. Extending adaptation to earlier or multiple evaluations would alter subsequent states and require on-policy student rollouts or synchronized teacher trajectories.

3.4 Parameter-Free Schedule Continuation

After endpoint alignment and CRLU, TrajScale has a target-grid clean latent

$$\hat{z}_s^H = U_\psi(\tilde{z}_s^L),$$

but the next denoising evaluation expects an intermediate state at the corresponding scheduler level. We therefore reconstruct

$$x_{s+1}^H = (1 - \sigma_{s+1})\hat{z}_s^H + \sigma_{s+1}\epsilon^H, \quad \epsilon^H \sim \mathcal{N}(0, I).$$

The noise is sampled directly on the target-resolution grid. This equation is a scheduler operation, not a learned module and not a third training objective. It completes the scale transition in the clean-latent domain and then initializes the target-grid state at the intended scheduler coefficient, rather than resizing a low-resolution noisy state or velocity field across grids.

Starting from x_{s+1}^H , the unmodified Wan denoiser performs the remaining evaluations in the high-resolution latent space. The suffix refines the transferred structure and uses the base model’s native prior to synthesize fine textures and spatial frequencies that are not recoverable from the low-resolution latent alone. State reconstruction merely enables this native continuation; the frozen high-resolution suffix, not the reconstruction operation itself, performs generative refinement.

The same procedure applies to conventional many-step and temporally distilled few-step samplers. For any valid transition index s , the sampler executes its prefix at low resolution, invokes EAA and CRLU once, reconstructs the next scheduler state on the target grid, and completes the remaining evaluations at high resolution. TrajScale neither changes the base scheduler nor retrains the frozen denoiser. Concrete transition indices and quality- or efficiency-oriented operating points are specified in Section 4.

4 Experiments

4.1 Experimental Objectives and Setup

Our evaluation addresses four questions: (RQ1) whether TrajScale establishes a useful end-to-end quality–efficiency

frontier; (RQ2) whether CRLU learns a more faithful cross-resolution operator than fixed interpolation; (RQ3) how EAA changes endpoint error, automatic quality dimensions, and human perceptual preferences when CRLU is deployed on an unconverged trajectory; and (RQ4) whether the same spatial transition remains compatible with a 4-step distilled sampler.

For the 50-step pipeline, we generate approximately 1,000 reference videos using Wan2.1 T2V-1.3B. Each video contains 81 frames at 720×1248 resolution and 16 fps. Bicubic RGB downsampling produces corresponding 480×832 and 368×640 videos, which are re-encoded by the frozen Wan VAE to construct aligned cross-resolution latent pairs. EAA training pairs consist of the low-resolution state immediately preceding a designated transition and the endpoint of the corresponding complete low-resolution trajectory.

The 4-step experiments use approximately 5,000 videos generated by Wan2.1 T2V-14B StepDistill-CfgDistill. CRLU is trained on $368 \times 640 \rightarrow 720 \times 1248$ clean latent pairs, while EAA pairs the state preceding step 3 with the complete step-4 low-resolution endpoint. Training, validation, and test partitions are separated by both text prompts and random seeds.

CRLU is optimized with AdamW for up to 50k steps using a learning rate of 1×10^{-4} , *bf16* precision, and an EMA decay of 0.9999. EAA is trained for up to 10k steps with AdamW, a learning rate of 5×10^{-5} , and LoRA rank/scale 32/32. Training uses four NVIDIA H100 GPUs and requires approximately 8 hours for CRLU and 33 hours for EAA. All efficiency measurements are averaged over five independent cold-start runs. Validation selects alignment intensity $\gamma = 0.75$ for the reported Wan50 and distilled operating points; the intensity sweep is analyzed in Section 5.2. Since all supervision is generated by the frozen target model, TrajScale requires neither external paired videos nor an additional upscaling teacher.

4.2 Baselines and Evaluation Protocol

System-level comparisons include **Native-HR Sampling**, which executes all denoising evaluations at 720p; **Trilinear Handoff**, which performs mixed-resolution inference using fixed trilinear latent interpolation; and **Post-Generation Latent Cascade**, which completes the low-resolution trajectory, applies CRLU, and performs one manually selected low-intensity high-resolution refinement evaluation. We additionally report **Native-LR Sampling** as a low-resolution reference. The cascade baseline tests the alternative of postponing scale conversion until generation has finished, when the high-resolution trajectory entry level must be chosen externally.

Module interactions are evaluated through a 2×2 diagnostic design that crosses CRLU input states {Unaligned, EAA-aligned} with resolution operators {Trilinear, CRLU}. The four configurations are Trilinear Handoff, EAA+Trilinear, CRLU-only, and the complete TrajScale framework. The comparison from Trilinear to CRLU-only isolates the learned upsampling effect; the comparison from CRLU-only to TrajScale isolates transition-time input alignment. EAA+Trilinear is retained as a diagnostic to show that endpoint calibration cannot

replace an accurate scale operator. Oracle input-domain variants and a joint transition mapper are reported only as diagnostic references in the Supplementary Material.

For endpoint alignment, we report endpoint latent L_1 , decoded PSNR and LPIPS, temporal latent differences, and sample-wise win rates. CRLU is evaluated using latent L_1 , PSNR, SSIM, LPIPS, and temporal L_1 . End-to-end quality is measured with VBench (Huang et al. 2024) and human evaluation. We define **VBench-5** as the unweighted mean of subject consistency, background consistency, motion smoothness, aesthetic quality, and imaging quality.

All paired configurations share identical prompts, random seeds, initial noise, schedulers, and guidance parameters. Prompt-level score differences are analyzed using 10,000 paired bootstrap iterations for 95% confidence intervals and two-sided exact sign tests. Human evaluation uses randomized, double-blind comparisons over 10 matched prompts, with three independent judges per prompt and prompt-level majority voting. Inter-rater agreement is reported in the Supplementary Material.

Efficiency measurements include end-to-end cold-start latency, peak GPU memory, low- and high-resolution denoising evaluations, and speedup over Native-HR Sampling. All EAA, CRLU, VAE, scheduler-state reconstruction, and dynamic model-loading overheads are included.

5 Results and Analysis

5.1 System-Level Quality–Efficiency Trade-off

Table 1 compares the standard 50-step Native-HR pipeline with the two principal TrajScale operating points. TrajScale-Q moves the first 40 evaluations to the low-resolution grid and retains 10 high-resolution suffix evaluations, reducing latency from 253.10 to 138.36 seconds ($1.83\times$). TrajScale-E retains only five high-resolution evaluations and further reduces latency to 114.26 seconds ($2.22\times$). Their VBench-5 scores are 0.80983 and 0.80792, compared with 0.82836 for Native-HR Sampling. These operating points show that later transitions increase acceleration while reducing the high-resolution budget available for native detail synthesis.

The Post-Generation Latent Cascade reaches 86.45 seconds ($2.93\times$) but decreases VBench-5 to 0.80093, with inferior scores on 9/10 prompts relative to Native-HR ($p = 0.02148$). Completing the low-resolution trajectory before scale conversion maximizes speed, but leaves only a manually selected low-noise refinement evaluation and therefore less capacity for structural and texture refinement. Peak memory remains approximately 26 GiB across Native-HR, TrajScale-Q, TrajScale-E, and the post-generation cascade, indicating that TrajScale primarily reduces runtime rather than peak memory.

5.2 Component Analysis

Cross-resolution latent upsampling. Figure 4(a) and Table 2 isolate the cross-resolution operator before considering transition-time mismatch. Over 50 paired clean latent samples, CRLU substantially outperforms trilinear interpolation. For $480p \rightarrow 720p$, it reduces latent L_1 from 0.1929 to 0.0991, a 48.6% reduction; improves decoded PSNR by 5.71

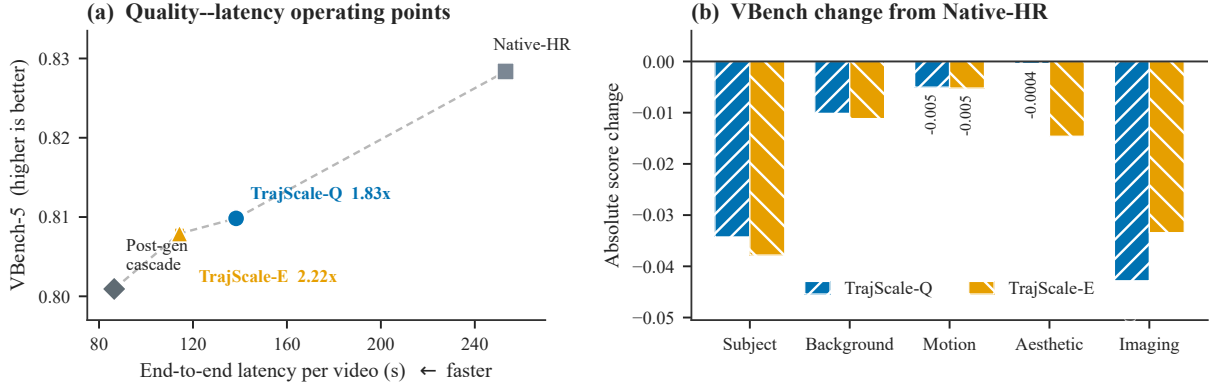


Figure 3: Quality--efficiency trade-offs relative to Native-HR Sampling. (a) TrajScale-Q and TrajScale-E provide quality- and efficiency-oriented operating points, while the post-generation latent cascade achieves higher speed at a larger quality cost. Error bars show the standard deviation of five cold-start latency runs. (b) Absolute per-dimension VBench changes relative to Native-HR over ten matched prompts.

Method	LR/HR evals.	Time (s) ↓	Latency red.	Speedup ↑	VBench-5 ↑	Retention ↑
Native-HR Sampling	0/50	253.10 ± 1.53	—	1.00×	0.82836	100.00%
TrajScale-Q (Step 40)	40/10	138.36 ± 0.50	45.33%	1.83×	0.80983	97.76%
TrajScale-E (Step 45)	45/5	114.26 ± 1.52	54.85%	2.22×	0.80792	97.53%

Table 1: System-level comparison against Native-HR Sampling. Timings average five cold-start runs, and quality is evaluated on ten matched prompts with all processing overheads included.

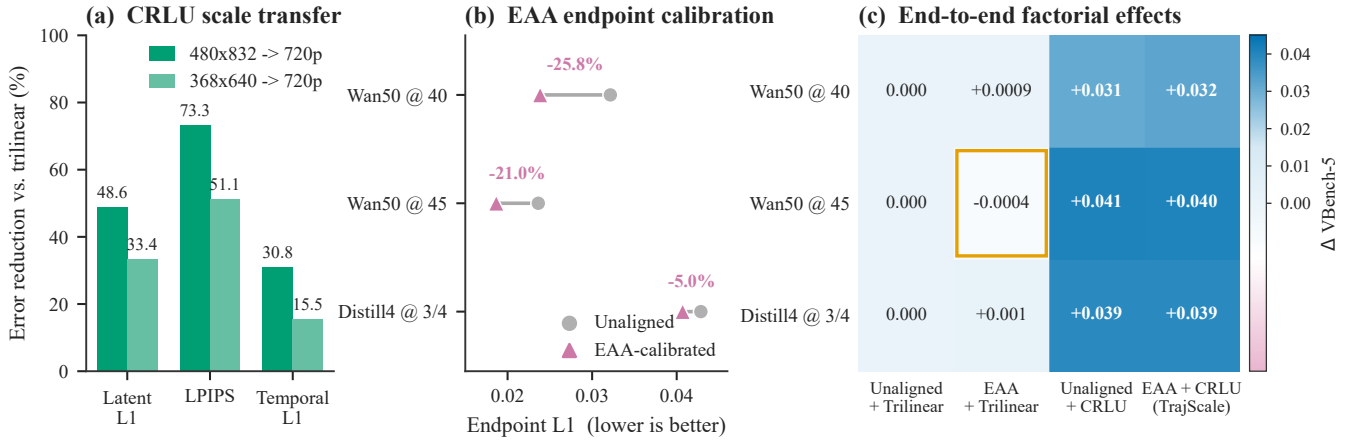


Figure 4: Evidence for the revised two-role decomposition. (a) CRLU reduces paired clean-latent errors relative to trilinear interpolation at both scale ratios. (b) EAA lowers the endpoint residual at the two Wan50 transitions and on the 4-step distilled sampler; circles and triangles denote unaligned and EAA-calibrated inputs. (c) End-to-end VBench-5 changes relative to unaligned trilinear handoff. CRLU produces the dominant aggregate gain, whereas EAA has a much smaller and transition-dependent marginal effect.

dB; reduces LPIPS from 0.2358 to 0.0629; and lowers temporal L_1 by 30.8%. CRLU wins on all 50 samples for PSNR, LPIPS, and temporal consistency, and on 49/50 samples for SSIM.

The more aggressive 368p $\rightarrow$ 720p transition shows the same trend: latent L_1 , LPIPS, and temporal L_1 decrease by 33.4%, 51.1%, and 15.5%, respectively. These improvements show that the primary cross-resolution capability comes from learning the VAE-specific latent correspondence rather than relying on a geometry-only resize rule.

Transition-time endpoint alignment. With the scale operator fixed, EAA tests how endpoint calibration changes the input presented to CRLU. At alignment intensity $\gamma = 0.75$, Figure 4(b) shows that EAA reduces residual low-resolution endpoint L_1 by 25.82% at step 40 and 21.03% at step 45. Both settings improve all 10 evaluation samples, with paired confidence intervals excluding zero. EAA also increases decoded PSNR by 2.55 and 2.23 dB and reduces latent temporal-difference errors by 23.44% and 21.24%, respectively. On the 4-step distilled model, it yields a smaller but consistent 5.03% endpoint-error reduction. All three comparisons have a 10/0 paired win rate and two-sided exact sign-test

Input $\rightarrow$ Output	Operator	Latent $L_1 \downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Temp. $L_1 \downarrow$
480p $\rightarrow$ 720p	Trilinear	0.1929	24.29	0.7159	0.2358	0.02121
480p $\rightarrow$ 720p	CRLU	0.0991	30.00	0.8696	0.0629	0.01468
368p $\rightarrow$ 720p	Trilinear	0.2444	23.42	0.6740	0.3352	0.02388
368p $\rightarrow$ 720p	CRLU	0.1628	24.10	0.7592	0.1640	0.02017

Table 2: CRLU versus trilinear interpolation on paired clean latents ($n = 50$).

$p = 0.00195$; full confidence intervals are reported in Supplementary Table S2.

Alignment-intensity validation is non-monotonic. The selected intensity $\gamma = 0.75$ produces the lowest endpoint error and strongest downstream quality among the tested settings, whereas weaker or full-strength correction is less effective. This result motivates treating EAA as a calibrated deployment control, not as an unconstrained replacement for the frozen predictor or as a module whose effect should grow monotonically with strength.

Hierarchical module effects. The 2×2 diagnostic in Figure 4(c) and Table 3 confirms the hierarchical roles of the two learned modules. Replacing trilinear interpolation with CRLU provides the principal broad quality gain, increasing VBench-5 by approximately 0.031–0.041 across the three pipelines. With CRLU fixed, EAA has a narrower transition-specific effect: it improves the imaging-quality dimension by +0.00599, +0.00254, and +0.00479 at step 40, step 45, and on the 4-step sampler, respectively. Because these detail-oriented gains primarily affect one component of the five-dimensional aggregate, they appear as smaller changes in averaged VBench-5.

The comparison supports the proposed causal chain. CRLU establishes the cross-resolution representation and supplies the dominant improvements in motion smoothness, aesthetics, and overall imaging quality. EAA does not replace this capability; it calibrates the transition-time input that CRLU receives, enriching localized textures and fine structures that may be underrepresented by aggregate automatic metrics. The remaining frozen high-resolution suffix then performs native generative refinement.

5.3 Perceptual Effects and Distilled-Sampler Compatibility

Human evaluation clarifies the effect that is diluted in the aggregate VBench-5 mean. With CRLU fixed, EAA receives fine-detail win/loss/tie counts of 8/2/0 at step 40 and 9/0/1 at step 45; overall-quality judgments are 6/2/2 and 6/0/4, respectively. The step-45 detail and overall preferences are significant under the exact sign test after ties are removed, whereas the step-40 results are directional but not significant. The evidence therefore supports a localized perceptual effect, not a claim that EAA uniformly improves every sample or metric.

The additional fine-scale content introduces a spatial-temporal trade-off: temporal stability favors the unaligned CRLU state in both settings. Together with the near-zero marginal VBench-5 changes in Table 3, this result sharpens the role of γ : EAA moves the operating point toward sharper local rendering, while smaller alignment preserves more of the unaligned trajectory dynamics. CRLU and the

remaining high-resolution suffix provide the broader spatial and temporal reconstruction in either case.

CRLU provides the broader perceptual improvement. At step 45, both CRLU-only and TrajScale-E obtain 10/10 prompt-level wins over trilinear handoff for fine details and overall quality, while also improving temporal stability. EAA adds a narrower, strength-dependent local-detail adjustment on top of this learned cross-resolution reconstruction.

TrajScale is also compatible with the 4-step distilled Wan sampler. On the 368p trajectory, Trilinear Handoff achieves a VBench-5 score of 0.81796, whereas TrajScale-D4 reaches 0.85680. Cold-start latency increases only from 167.09 to 170.29 seconds. As in the 50-step setting, CRLU supplies the primary cross-resolution quality improvement, while EAA reduces the transition-time endpoint residual and improves the imaging-quality component. This result supports the orthogonality of spatial mixed-resolution sampling and temporal step distillation.

Qualitative comparisons in Figure 1(a) further illustrate CRLU’s recovery of high-frequency textures and EAA’s localized adjustments. Extended spatial crops and temporal sequences are provided in Figure S1. The content-matched Full-LR endpoint reference is produced from a complete 368p trajectory and is used only for frame-level structural comparison; true 720p Native-HR Sampling is used for the quantitative evaluation in Table 1.

6 Discussion and Limitations

TrajScale separates cross-resolution representation learning from transition-specific input calibration. CRLU transfers structure and motion to the target latent grid, EAA controls how an unfinished low-resolution prediction is presented to this learned mapping, and the frozen high-resolution suffix completes underdetermined detail synthesis. Scheduler-consistent state reconstruction is a parameter-free bridge between the clean-latent mapping and the native suffix; it enables, rather than independently performs, high-resolution refinement. This hierarchy matches the empirical results: CRLU provides the broad quality gain, while endpoint metrics, imaging-quality changes, and human preferences show that EAA controls a narrower transition-time failure mode. EAA’s small aggregate effect and opposing temporal preferences are not contradictions; they expose the detail–stability trade-off hidden by a single averaged score.

Both learned modules use model-endogenous supervision. The frozen Wan generator supplies high-resolution videos for paired CRLU latents and complete low-resolution trajectories for EAA, avoiding external paired datasets, standalone super-resolution teachers, or modification of the base denoiser. The two supervision streams are nevertheless not identical distributions. EAA reduces the residual between

Sampler / transition	Trilinear	EAA+Tri.	CRLU-only	TrajScale	Upsampling effect	Alignment effect	Full effect
Wan50 / Step 40 (TrajScale-Q)	0.77812	0.77901	0.80896	0.80983	+0.03085	+0.00087	+0.03171
Wan50 / Step 45 (TrajScale-E)	0.76776	0.76734	0.80842	0.80792	+0.04066	-0.00050	+0.04016
Distill4 / Step 3-of-4	0.81796	0.81909	0.85670	0.85680	+0.03874	+0.00010	+0.03884

Table 3: End-to-end factorial results on VBench-5 ($n = 10$). “Alignment effect” denotes the marginal effect of enabling EAA with CRLU fixed.

Comparison	Details	Overall	Temporal
EAA vs. Unaligned, CRLU (S40)	8/2/0	6/2/2	1/9/0
EAA vs. Unaligned, CRLU (S45)	9/0/1	6/0/4	0/8/2
CRLU-only vs. Trilinear (S45)	10/0/0	10/0/0	8/0/2
TrajScale-E vs. Trilinear (S45)	10/0/0	10/0/0	9/0/1

Table 4: Prompt-level majority votes (W/L/T) over ten prompts with three raters per prompt. Bold entries have exact sign-test $p < 0.05$.

an intermediate prediction and its native low-resolution endpoint, but does not eliminate the remaining gap between native endpoints and the RGB-downsampled paired latents used to train CRLU.

Several limitations remain. CRLU is tied to the target VAE, generator distribution, resolution ratio, and downsampling process, so cross-model and cross-degradation transfer remain unverified. Fixed handoff steps require step-specific EAA parameters and a validated alignment intensity; timestep-conditioned or jointly calibrated adapters would be needed for adaptive transitions. Quality also depends on the manually selected suffix length and on whether the scheduler-state reconstruction remains appropriate for other solver families. Finally, TrajScale addresses model-generated in-trajectory spatial scaling rather than real-world degradations such as compression, blur, or sensor noise, and inherits the base generator’s content risks and safety requirements.

7 Conclusion

We presented TrajScale, a learned in-trajectory latent up-sampling framework for efficient high-resolution video generation. CRLU establishes a model-specific cross-resolution clean-latent mapping, EAA corrects the input mismatch created when that mapping is deployed before low-resolution convergence, and the frozen high-resolution suffix synthesizes details that are underdetermined by the low-resolution input. Both learned modules are trained from supervision generated by the frozen target model, and the base denoiser and scheduler remain unchanged.

TrajScale-Q and TrajScale-E achieve $1.83\times$ and $2.22\times$ speedups while retaining 97.76% and 97.53% of Native-HR VBench-5 performance. CRLU consistently outperforms trilinear interpolation and supplies the primary cross-resolution quality gain. EAA reduces endpoint residuals and provides a controllable local-detail adjustment, but its small aggregate quality effect and temporal-stability trade-off make it a deployment calibrator rather than a second general-purpose quality module. These results show that learned scale transfer, transition-specific calibration, parameter-free schedule initialization, and native suffix refinement provide a practical division of labor for mixed-resolution video diffusion.

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